
Hurricane Tracking with the Kalman Filter

<https://github.com/sardorsob/Hurricanepaths>

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Abstract

Hurricane track forecasting is a sequential estimation problem under substantial uncertainty: observations are noisy, atmospheric dynamics are partially observed, and errors in early predictions compound into large positional deviations at longer lead times. Rather than treating forecasting as a deterministic regression task, we frame it as a filtering problem, modeling each storm’s true position and velocity as a latent state evolving according to linear-Gaussian dynamics and performing exact sequential Bayesian inference via the Kalman filter. Applied to the IBTrACS global cyclone database - 722,040 observations across 13,530 storms - the baseline filter achieves a filtering-mode RMSE of 26.85 km and a mean forecast error of 11.9 km at 6 hours, with errors growing rapidly beyond 48 hours as nonlinear atmospheric steering dominates. We further evaluate a feature-adaptive variant that modulates process noise according to storm covariates; innovation diagnostics reveal that none of the candidate features reliably predict filter uncertainty ($R^2 < 0.10$), explaining the negligible gains the adaptive model produces. Our results establish both the promise and the hard limits of position-only linear filtering for this problem.

Keywords: Kalman filter, sequential Bayesian inference, state-space model, probabilistic forecasting, tropical cyclone, IBTrACS, uncertainty quantification

1 Introduction and Motivation

Accurate hurricane track forecasting is *the* most operationally consequential output of tropical meteorology. Behind evacuation orders, pre-positioning of emergency resources, and coastal shelter-in-place decisions lies a single number: where will the storm be, and when. Even modest positional errors propagate directly into those decisions, and the cost of a missed forecast falls disproportionately on those with the least time to respond [11]. The core difficulty is that hurricane motion is not deterministic. It emerges from complex, partially observed atmospheric interactions, and even the highest-quality observational records carry substantial noise and storm-to-storm heterogeneity [13]. A deterministic model that ignores this uncertainty doesn’t just produce point estimates - it produces overconfident ones, with no principled way to communicate how wrong those estimates might be.

What we actually want is a model that reasons explicitly about what it doesn’t know. A natural framing is to treat the storm’s true state - its position and velocity - as a latent variable, never directly observed, only estimated from noisy measurements. Early work by Brill demonstrated that

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this filtering perspective, implemented via the Kalman filter, could propagate uncertainty explicitly through a storm’s trajectory, correcting noisy observations with a physically motivated motion model [8]. Building on this foundation, we develop a Kalman filter-based probabilistic framework applied to a comprehensive global cyclone database, asking whether such a model can improve both the accuracy of hurricane position estimates and the calibration of uncertainty over a storm’s lifetime. The goal is a system that does not merely predict where a storm will go, but honestly represents how confident we should be in that prediction - and where the limits of that confidence lie.

In Section 2, we describe the IBTrACS dataset, how we handled missing data, and the features we engineered from raw storm observations. In Section 3, we formalize the Kalman filter as a linear-Gaussian state-space model, define the inference procedure, and describe our baseline and feature-adaptive variants. Section 4 presents forecast accuracy across lead times, distributional diagnostics, and the comparison between models. We conclude in Section 5 with a summary of findings, the fundamental limitations of position-only linear filtering, and directions for future work.

2 Data

This study draws on the IBTrACS (International Best Track Archive for Climate Stewardship) dataset, produced by NOAA’s National Centers for Environmental Information [3]. IBTrACS is the most comprehensive publicly available record of global tropical cyclone activity, spanning 1842 to 2025. We structured the data as a panel of storm observations, covering 722,040 time steps across 13,530 unique storms. Within each storm i at time step t , we used the following:

1. Position (latitude/longitude), converted to local Cartesian coordinates $(x_{\text{km}}, y_{\text{km}})$: $\mathbf{y}_{it}$
2. Derived velocity components $(vx_{\text{km}}, vy_{\text{km}})$ over six-hour intervals: $\mathbf{v}_{it}$
3. Track curvature, defined as the rate of directional change between consecutive velocity vectors: κ_{it}
4. Binary land-proximity indicator, flagging observations within 200 km of a major landmass: L_{it}
5. Latitude band and intensity category, one-hot encoded: $\mathbf{z}_{it}$

We excluded storms with only a single observation, as velocity cannot be computed without at least two consecutive positions - this removed 80 storms from the dataset. Remaining gaps were treated as random measurement failures unrelated to any particular variable.

The dataset reports storm speed and direction directly, but because we could not verify the accuracy or internal consistency of those measurements, we derived velocity from consecutive position observations instead. Positions in IBTrACS are recorded in latitude and longitude, which cannot be treated as Euclidean coordinates without introducing distortions that accumulate over a storm’s track. To address this, we anchored each storm’s trajectory at its first observation, treating it as the origin of a local flat coordinate system, and used haversine distance calculations to decompose each subsequent displacement into east–west and north–south components in kilometers. Velocities were computed over six-hour intervals rather than three-hour intervals to improve numerical stability.

The engineered features in items 3–5 above were motivated by known physical drivers of storm motion. Track curvature captures the tendency of storms to recurve, with extreme values clipped to prevent distortion during parameter estimation. The land-proximity indicator reflects the well-documented disruption to storm dynamics near coastlines [8]. Latitude bands encode the large-scale steering regime: tropical storms follow trade-wind-driven westward tracks, while higher-latitude storms tend to recurve poleward before turning east under the mid-latitude westerlies [1]. Intensity category was one-hot encoded because the dynamical regime of a storm - from tropical depression to major hurricane - influences how it responds to its environment.

Two structural features of the dataset bear directly on modeling choices. First, a small number of older storms predating satellite coverage were recorded at irregular time intervals; we flagged these for potential exclusion, as the Kalman filter assumes fixed six-hour time steps. Second, there is pronounced geographic imbalance: the Western Pacific accounts for roughly 241,000 observations while the South Atlantic contains only 119. This disparity motivates either region-specific modeling or observation weighting to prevent the dominant basin from overwhelming parameter estimation.

3 Methodology

Rather than treating hurricane track forecasting as a regression problem - mapping features to a point estimate of future position - we frame it as a sequential estimation problem. At every time step, we maintain a full probability distribution over where the storm is and how fast it is moving, updating that distribution as each new noisy observation arrives. The Kalman filter is the natural tool for this: a continuous-state Hidden Markov Model that computes $\mathbb{P}(x_t \mid y_{1:t})$ - the posterior over the current state given all observations up to time t - exactly, via matrix operations, because the model is linear and the noise is Gaussian [15, 17]. The analog to the discrete HMM forward algorithm is exact here rather than approximate, because Gaussians are closed under linear transformations.

3.1 State Representation

We represent each hurricane as a sequence of hidden states x_t , where each state is a four-dimensional vector

$$x_t = (x_{\text{km}}, y_{\text{km}}, vx_{\text{km}}, vy_{\text{km}}), \quad (1)$$

comprising the storm’s east–west and north–south displacement from its first observed position, and the corresponding velocity components over six-hour intervals. The state is never observed directly - only a noisy two-dimensional position measurement $y_t \in \mathbb{R}^2$ is available at each step, making the velocity components permanently latent. The dependency structure follows the Markov property: x_{t+1} depends only on x_t , and y_t depends only on x_t , exactly the graphical structure of an HMM.

3.2 Model Specification

The system is governed by two equations. The *state transition model*,

$$x_{t+1} = A x_t + w_t, \quad w_t \sim \mathcal{N}(0, Q), \quad (2)$$

encodes constant-velocity dynamics: the transition matrix A updates position by adding the current velocity while holding velocity constant between steps. The process noise w_t with covariance Q captures everything the constant-velocity assumption misses - accelerations, decelerations, and turns. The *observation model*,

$$y_t = H x_t + v_t, \quad v_t \sim \mathcal{N}(0, R), \quad (3)$$

maps the full hidden state to the observed (noisy) position, where $H \in \mathbb{R}^{2 \times 4}$ selects only the position components of x_t . The measurement noise v_t with covariance R encodes the inherent uncertainty in the best-track position estimates.

A and H are fixed by physics and measurement structure and are not learned from data. The learned parameters are Q and R , both estimated from approximately 200 training storms: we ran the filter with initial guesses, recorded the prediction residuals at each step, and took their sample covariance - a form of maximum likelihood estimation over the residual distribution. This yielded R values of approximately 265 km² east–west and 169 km² north–south, and a large Q , reflecting the frequency with which storms violate the constant-velocity assumption through turns and intensity changes.

We implemented two model variants. The baseline uses a constant Q across all storms and all time steps. The feature-adaptive variant scales Q by a factor $s \in [1.0, 3.0]$ at each step as a function of track curvature, land proximity, and motion regime - increasing process uncertainty when the storm is turning sharply or approaching a coastline. The scaling parameters were selected on a held-out validation set via grid search, retaining the configuration that minimized positional error.

3.3 Inference and Forecasting

Inference proceeds via exact filtering [12]. At each time step the algorithm alternates between two operations. The *predict step* propagates the previous posterior forward through the transition model,

$$\mathbb{P}(x_t \mid y_{1:t-1}) = \mathcal{N}(A\hat{x}_{t-1}, AP_{t-1}A^\top + Q),$$

forming the one-step-ahead predictive distribution. The *update step* incorporates the new observation y_t via the Kalman gain K_t ⁵, correcting the mean and shrinking the covariance to yield the refined

⁵ $K_t = P_t^- H^\top (H P_t^- H^\top + R)^{-1}$, a matrix that optimally weights the new observation against the prior prediction as a function of the relative magnitudes of Q and R .

posterior $\mathbb{P}(x_t \mid y_{1:t})$. Because linearity and Gaussianity are preserved throughout, every posterior is representable exactly as a mean vector and a covariance matrix - the covariance matrix being the filter’s explicit, propagated account of its own uncertainty.

For evaluation, we used a sliding-origin cross-validation protocol: at each origin time t_0 , we ran the filter up to t_0 to obtain $\mathbb{P}(x_{t_0} \mid y_{1:t_0})$, then iterated the transition model forward without assimilating further observations to generate multi-step forecasts at lead times of 6, 12, 24, 48, and 72 hours. Uncertainty grows monotonically as Q accumulates across steps, the natural behavior of an honest probabilistic forecast under increasing temporal horizon.

4 Results

We evaluated both model variants using a sliding-origin cross-validation protocol across 2,680 held-out test storms, computing track errors at standard NHC lead-time marks [9]. Table 1 reports forecast accuracy across lead times.

Table 1: Track Forecast Error Summary Statistics

Lead Time	Mean Error (km)	RMSE (km)
6 hours	11.90	15.86
12 hours	24.53	34.50
24 hours	58.13	83.79
48 hours	158.95	216.74
72 hours	286.08	379.35

Short-term performance is competitive: error roughly doubles every twelve hours through the 24-hour mark, consistent with linear accumulation of process noise under constant-velocity dynamics [15]. Beyond 48 hours, degradation accelerates as recurvature and shifting steering flows produce systematic deviations the linear transition model cannot anticipate [10]. In filtering mode - where observations are assimilated continuously rather than withheld - the model achieves a mean error of 21.13 km and RMSE of 26.85 km, establishing the practical ceiling of what the model class can recover given continuous access to observations. The cumulative RMSE distribution (Appendix A, Figure 1) shows a median of 23.4 km and a 90th percentile of 43.4 km; the long right tail is driven by a small number of structurally difficult storms rather than systematic model failure.

The aggregate metrics, however, mask a more important finding. The forecast errors deviate substantially from the Gaussian assumptions the Kalman filter relies on [8, 15]. Appendix A, Figure 2 shows that at every lead time, the empirical error distribution is sharply concentrated near zero with a heavy right tail - a profile the fitted Gaussian consistently fails to capture, overestimating probability mass at moderate errors while underestimating both the near-zero peak and the far tail. This discrepancy compounds with lead time: by 48 hours, the median error is 111 km against a mean of 159 km, indicating that most forecasts are reasonable but a subset of large errors inflates the aggregate. The practical consequence is that the Kalman-derived covariance matrices understate tail risk [17] - forecast cones built from them will systematically undercover the true spread of possible tracks.

The innovation scatter (Appendix A, Figure 3) corroborates this picture. Residuals are centered near zero and symmetric across both coordinate directions, confirming that the constant-velocity transition model is unbiased and introduces no directional drift [12]. The cloud is nonetheless distinctly heavy-tailed: the filter is correctly centered, but systematically overconfident about the magnitude of its own uncertainty.

Table 2: Innovation Ratio Diagnostics and Predictive Power

Feature	High/Low Innovation ² Ratio	Predictive Power
Track Curvature	~ 1.0	Weak
Land Proximity	~ 1.0	Weak
Motion Regime	~ 1.0	Weak
Latitude Regime	~ 1.0	Weak

Table 2 presents a key diagnostic for the adaptive-Q experiment before we run it. For each candidate feature, we computed the ratio of mean squared innovation between periods when the feature is high versus low. Across all four - track curvature, land proximity, motion regime, and latitude regime - this ratio is approximately 1.0, and regression models predicting innovation variance from these features achieve $R^2 < 0.10$. None of the covariates reliably identify when the filter is about to make a large error. The implication is direct: modulating Q according to a feature that carries no information about filter uncertainty cannot improve forecast accuracy, regardless of how the scaling factors are set.

Table 3: Baseline vs. Adaptive-Q RMSE Across Lead Times

Lead Time	Baseline RMSE (km)	Adaptive-Q RMSE (km)	Improvement
6 hours	15.86	15.58	1.8%
12 hours	34.50	34.05	1.3%
24 hours	83.79	83.04	0.9%
48 hours	216.74	216.74	0.0%
72 hours	379.35	379.50	-0.04%

Table 3 confirms the prediction. Adaptive-Q yields sub-kilometer improvements at short lead times and no discernible benefit beyond 48 hours. Appendix A, Figure 4 makes the null result concrete: the RMSE curves for both models are nearly coincident at every lead time, and the pointwise error scatter lies almost entirely on the diagonal, indicating that individual forecasts are effectively unchanged by the adaptive mechanism. The failure is not a tuning artifact - it is a logical consequence of the innovation analysis.

Beyond the baseline-versus-adaptive comparison, the results reveal a meaningful relationship between observational history and filter performance. Appendix A, Figure 5 shows that storms with fewer than ten observations exhibit the highest and most variable RMSE, while both mean error and variance decline steadily as track length grows, tightening considerably beyond thirty observations. This is the expected behavior of a sequential Bayesian estimator: posterior uncertainty contracts as more observations are assimilated, and the filter’s state estimates stabilize. The practical implication is that forecast confidence should be conditioned on track length - projections for nascent or short-lived storms warrant wider uncertainty bands than the filter’s covariance matrices alone would suggest.

Finally, the adaptive-Q pipeline required five to six hours of computation, driven by per-step feature extraction at each origin. Given that the accuracy gains were negligible, this overhead is difficult to justify. The baseline filter’s cost, by contrast, is dominated by fixed-size matrix operations that scale gracefully with the number of storms. The bottleneck is not computation or data - it is the explanatory power of the features used to drive adaptation.

5 Discussion

This work developed a sequential Bayesian state-space framework for probabilistic hurricane track forecasting, framing the problem as one of latent state estimation rather than regression. The Kalman filter, applied to 13,530 IBTrACS storms, produces competitive short-term forecasts - mean errors below 60 km through 24 hours and a filtering-mode RMSE of 26.85 km under continuous observation - while maintaining an explicit, propagated account of its own uncertainty at every time step. This is the core advantage of the probabilistic framing: the model does not merely predict where a storm will go, it produces a full distribution over possible positions, updated in real time as new observations arrive.

The results also establish clearly where the model class breaks down. Forecast errors grow rapidly beyond 48 hours as nonlinear atmospheric dynamics - recurvature, shifts in steering flow, interactions with large-scale pressure systems - produce deviations the linear transition model cannot anticipate. The error distributions are right-skewed at every lead time, revealing that the Gaussian covariance matrices the filter produces understate tail risk. The filter is correctly centered but systematically overconfident about the magnitude of its own uncertainty. And the adaptive- Q experiment, which attempted to modulate process noise according to storm covariates, confirms that the bottleneck is not a tuning problem: none of the candidate features carry meaningful information about when the filter will make a large error, so no scaling of Q can address the underlying misspecification.

5.1 Limitations

The most fundamental limitation is the linearity assumption. The Kalman filter treats hurricane motion as a constant-velocity process perturbed by Gaussian noise - a reasonable approximation over short intervals during which the steering environment changes slowly, but an inadequate one over longer horizons. Storms recurve, accelerate, and interact with jet streams, shifting pressure gradients, and mid-latitude westerlies in ways that produce genuinely nonlinear trajectories [4, 6]. No choice of Q , adaptive or otherwise, can compensate for a transition model that is misspecified at the level of functional form.

Compounding this is an information-theoretic ceiling that applies to the entire model class. Hurricane tracks are steered by atmospheric conditions that evolve independently of the storm itself [4]. Our model conditions only on past positions and velocities - it has no access to the large-scale atmospheric state that actually determines where a storm will go. Accurate multi-day forecasting would require that information, and the IBTrACS record does not provide it. This ceiling cannot be raised by better parameter estimation or more data; it requires a fundamentally different observation model.

The observational record introduces additional uncertainty that is difficult to fully absorb. Best-track positions are post-analysis estimates rather than direct measurements, the six-hour sampling interval renders rapid trajectory changes invisible to the filter [9], and the dataset spans nearly two centuries of heterogeneous data quality - pre-satellite records carry substantially higher measurement uncertainty than modern observations, yet a single R matrix must represent both. Finally, errors compound across forecast steps: a small positional error at t becomes the origin of the prediction at $t + 1$, and by 48–72 hours the accumulated uncertainty renders the forecast unreliable for operational purposes.

5.2 Future Work

Several directions follow naturally from these limitations. The most direct extension is to move beyond linear-Gaussian dynamics. An Extended or Unscented Kalman Filter [14, 16] would permit nonlinear transition functions that could model recurvature more faithfully, and Gaussian Process regression [7] would provide a fully nonparametric probabilistic alternative without committing to a parametric motion model. Both approaches preserve the probabilistic framing while relaxing the assumptions that constrain the current model.

On uncertainty quantification, the right-skewed error distributions point to a specific fix: the Gaussian forecast cones the filter produces need to be replaced with calibrated, heavy-tailed alternatives. Ensemble forecasting with multiple initializations [2], combined with prediction intervals that adaptively widen based on historical error variance in similar storm regimes, would produce more honest uncertainty estimates - particularly important in operational contexts where the cost of underestimating tail risk is asymmetric.

Two data-quality interventions also seem worth pursuing. Training separate parameter sets for pre- and post-satellite eras - or downweighting older observations during covariance estimation - would reduce the influence of high-uncertainty historical records on the learned Q and R . Separately, spline interpolation [5] between six-hour observations would create a denser measurement sequence, though the uncertainty introduced by interpolation would itself need to be propagated through the filter rather than treated as ground truth. Finally, a synthetic evaluation using tracks generated from known parameters would provide a ground-truth benchmark that real-data evaluation cannot: it would allow us to assess directly how well the filter recovers its own assumptions, and at what sample sizes [8].

We used Gen AI for coding help and debugging mostly, and we verified its outputs by observing if the discussion worked or not. We also used it for research purposes with respect to possible applications in hurricane forecasting, and we verified its outputs by asking it to provide its sources and its recommendations, and validating those aforementioned sources.

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6a29d3d38474e9c62b746fa646899552_MIT6_438F14_Lec13.pdf.

Appendix A

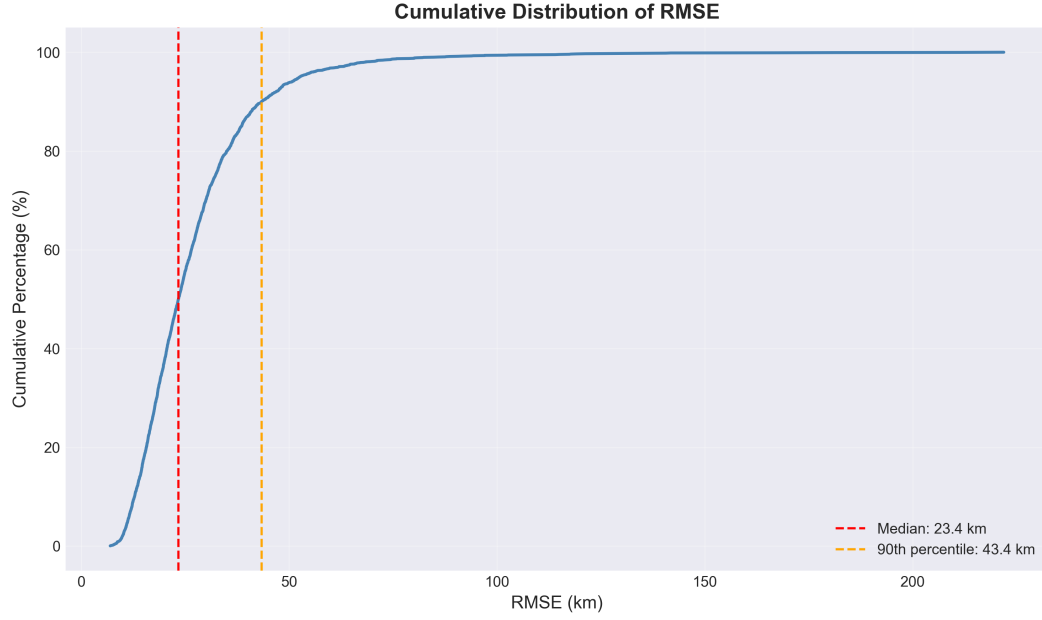


Figure 1: Cumulative distribution of track forecast RMSE across all forecast instances. Vertical lines mark the median, 75th percentile, and maximum error, summarizing the tail behavior of the baseline Kalman filter.

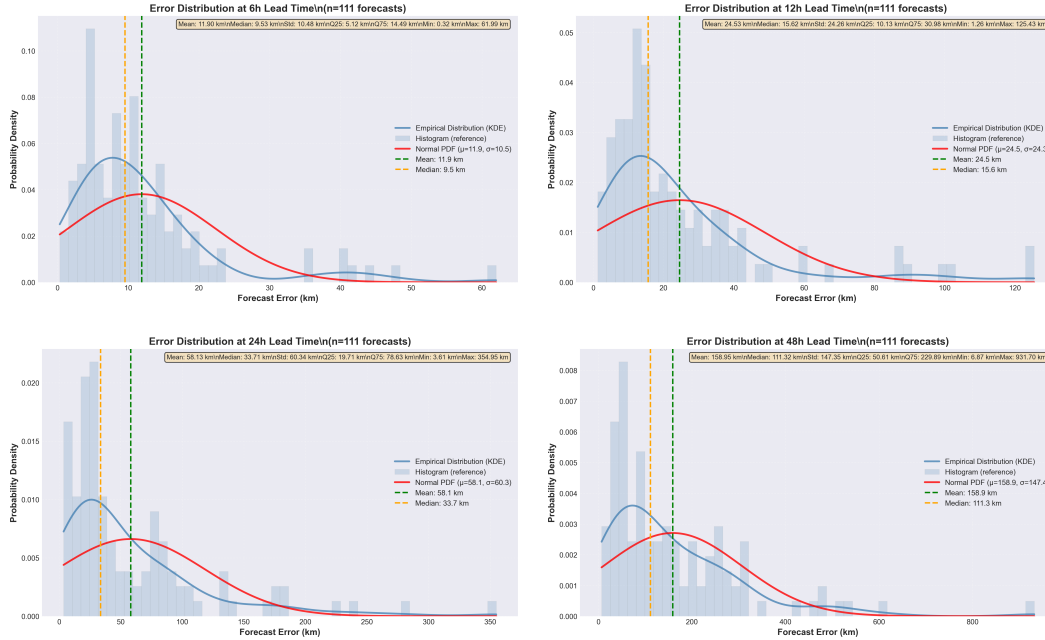


Figure 2: Forecast error distributions at four lead times (6, 12, 24, and 48 hours). Histograms, kernel density estimates, and fitted Gaussian curves highlight the increasing skewness and heavy right tails with longer lead time.

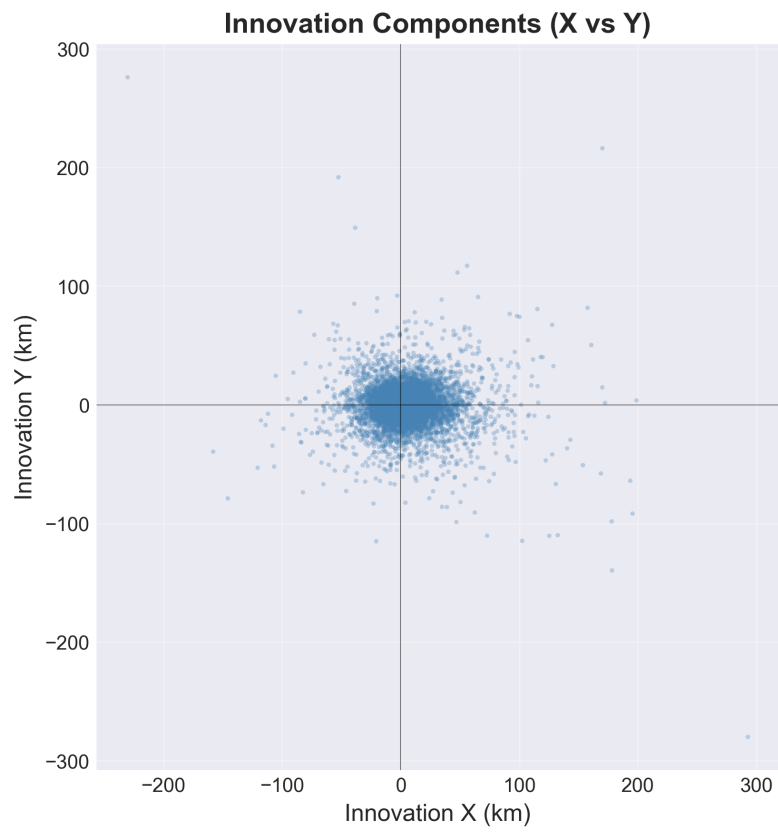


Figure 3: Scatter of innovation components in the along-track (X) and cross-track (Y) directions for all analysis times. The roughly circular, centered cloud indicates unbiased but heavy-tailed innovations.

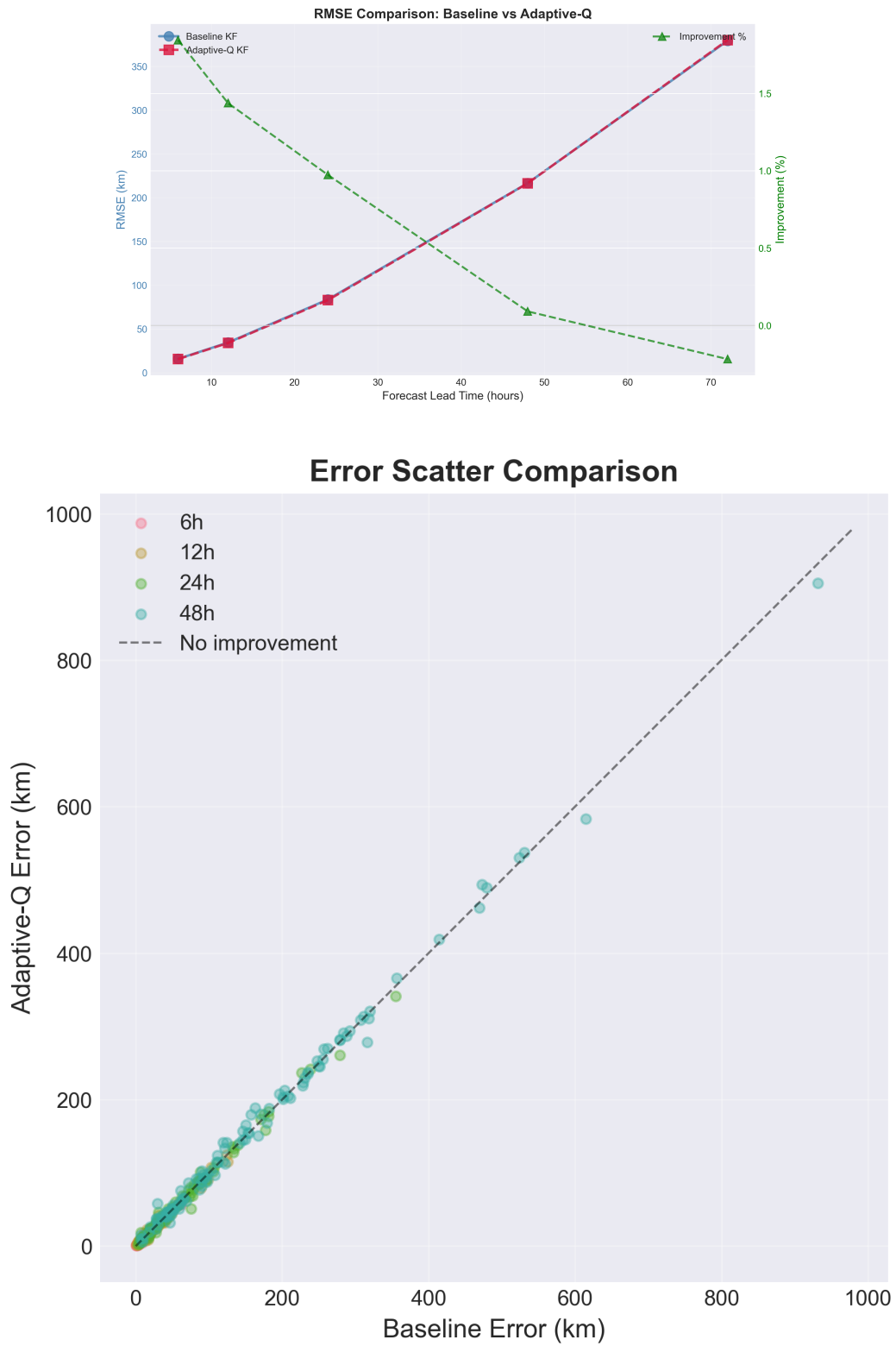


Figure 4: Comparison of baseline and Adaptive-Q Kalman filters. The upper panel shows RMSE by lead time, while the lower panel compares pointwise forecast errors, highlighting regimes where Adaptive-Q reduces large outliers.

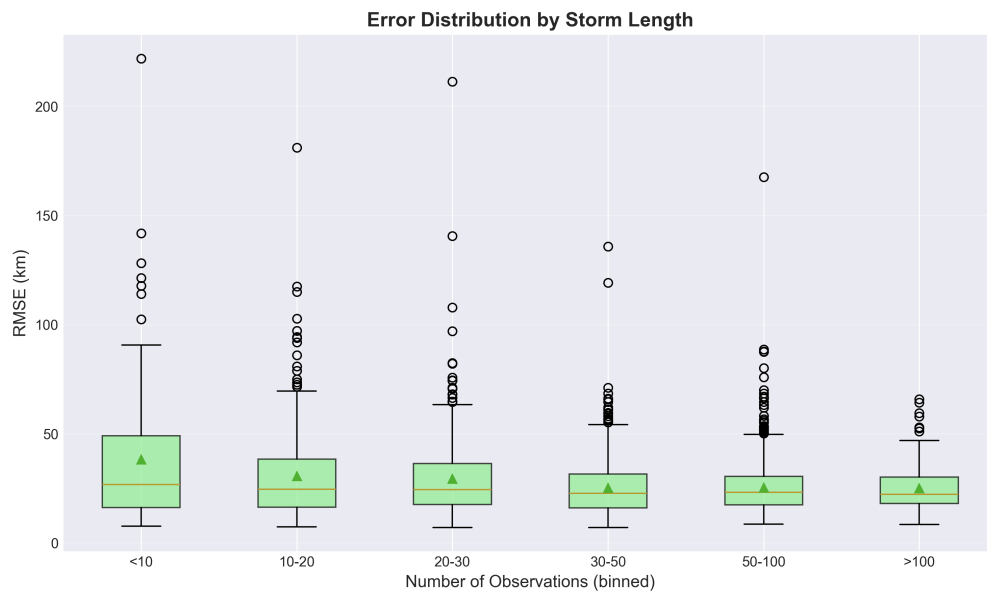


Figure 5: Relationship between track forecast error and storm length. The plot shows how storms with short observational histories tend to exhibit higher forecast error, reflecting limited information for estimating initial motion and dynamical regime. Longer storms generally show reduced error variance as the filter accumulates more observations and stabilizes state estimates.